

Curve fitting

1. Find a parabola that *exactly* passes through the three points $(0, 4)$, $(1, 9)$, and $(4, -2)$ by constructing a linear equation and then solving it.

Use the three-parameter model $y(x) = \sum_{i=1}^3 \beta_i f_i(x)$, where the basis functions are $f_i(x) = x^{i-1}$. Then construct the matrix with components $(M)_{ij} = f_j(x_i)$, so

$$M = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 4 & 16 \end{pmatrix}.$$

We'd like $y(x_i) = \sum_j \beta_j (M)_{ji} = y_i$, so the equation to solve is

$$M\boldsymbol{\beta} = \begin{pmatrix} 4 \\ 9 \\ -2 \end{pmatrix}.$$

The solution is

$$\boldsymbol{\beta} = \frac{1}{6} \begin{pmatrix} 24 \\ 43 \\ -13 \end{pmatrix},$$

which means that the desired polynomial is described by

$$y(x) = 4 + \frac{1}{6}(43x - 13x^2).$$

2. Various voltages v_j are applied to a resistor (a device contained in almost all electronic circuits) and the resulting currents i_j are measured:

v [V]	i [mA]
0.20	27
0.40	49
0.60	64
0.80	85
1.0	108

- (a) The resistance R should satisfy Ohm's law, $v = iR$. However, there may be a systematic offset in your voltmeter. In order to take this into account, instead apply a least-squares fit to the model $v = v_{\text{off}} + iR$. Find both v_{off} and R , assuming that the voltage measurements are much more accurate than the current measurements. [Note that you might want to use a calculator or a computer here.]

We want to fit to the model $i(v) = (v - v_{\text{off}})/R$. If the two basis functions are $i_1(v) = 1$ and $i_2(v) = v$, the two free parameters are thus

$$\boldsymbol{\beta} = \begin{pmatrix} -v_{\text{off}}/R \\ 1/R \end{pmatrix}.$$

From lecture, we want to solve

$$M^T M \boldsymbol{\beta} = M^T \mathbf{i},$$

where $\mathbf{i}$ is the vector of current measurements and

$$M = \begin{pmatrix} 1 & v_1 \\ \vdots & \vdots \\ 1 & v_5 \end{pmatrix} = \begin{pmatrix} 1 & 0.2 \\ 1 & 0.4 \\ 1 & 0.6 \\ 1 & 0.8 \\ 1 & 1.0 \end{pmatrix}.$$

Also,

$$M^T M = \begin{pmatrix} 5.0 & 3.0 \\ 3.0 & 2.2 \end{pmatrix},$$

and (rounding to two significant figures)

$$M^T \mathbf{i} = \begin{pmatrix} \sum_{j=1}^5 i_j \\ \sum_{j=1}^5 v_j i_j \end{pmatrix} = \begin{pmatrix} 330 \\ 240 \end{pmatrix}.$$

We must therefore solve

$$\begin{pmatrix} 5.0 & 3.0 \\ 3.0 & 2.2 \end{pmatrix} \begin{pmatrix} -v_{\text{off}}/R \\ 1/R \end{pmatrix} = \begin{pmatrix} 330 \\ 240 \end{pmatrix}.$$

Doing so gives

$$\begin{pmatrix} -v_{\text{off}}/R \\ 1/R \end{pmatrix} = \begin{pmatrix} 7.2 \\ 99 \end{pmatrix}$$

Hence,

$$R = 1/99 \text{ k}\Omega = 10 \Omega, \quad v_{\text{off}} = -(7.2 \text{ mA})(10 \Omega) = -72 \text{ mV}.$$

- (b) Ohm's law is an idealization. Suppose that you want to check for corrections to this law by introducing a (large) "nonlinearity scale" in the form of a new constant v_{nl} :

$$v + v^2/v_{\text{nl}} = v_{\text{off}} + iR.$$

Explain how you could use fitting to try to measure the voltage scale v_{nl} at which Ohm's law begins to fail (without actually applying very large voltages to the resistor). Which equations would you try to solve?

Fit to the model equation

$$i = \frac{1}{R}(-v_{\text{off}} + v + v^2/v_{\text{nlín}}).$$

We can choose the basis functions to be $i_1(v) = 1$, $i_2(v) = v$, and $i_3(v) = v^2$, so

$$M = \begin{pmatrix} 1 & v_1 & v_1^2 \\ \vdots & \vdots & \vdots \\ 1 & v_n & v_n^2 \end{pmatrix}.$$

Then,

$$M^T M \begin{pmatrix} -v_{\text{off}}/R \\ 1/R \\ 1/(v_{\text{nlín}}R) \end{pmatrix} = M^T \mathbf{i}.$$

Solving this would allow $v_{\text{nlín}}$ to be read off.

Note however that one important issue not addressed in this module is the uncertainty in the fit parameters. It would be easy in practice to calculate a $v_{\text{nlín}}$ which is completely unreliable. More care is required when analyzing real data.